

ER Project Dashboard Report

Executive Summary:

This ER hospital dashboard focuses on analyzing emergency room operations to drive measurable business value through improved patient experience, operational efficiency and revenue optimization. The primary goal is to optimize ER operations to increase patient volume, reduce average wait times, increase patient satisfaction scores and optimize staffing allocation during peak periods. Based on analysis, weekends into Mondays have high patient volumes with peak times from 10pm - 5am with a potential correlation with extended wait times as a result and lowered patient satisfaction scores. Interestingly, neurology, physiology, gastroenterology and cardiology department referrals accounted for some of the fewest patient visits yet had the majority of the highest wait times. Lastly women, individuals of mixed ethnicity and the elderly had the lowest patient satisfaction. In light of these findings, the ER management team can make informed decisions to meet the desired targets. This visualization is based on synthetic ER data meant to simulate real-world patient records observed over 2 years and can be filtered through the specific years, either 2019 or 2020 as well as gender. In addition, this dashboard is static, as further data will not be added to the model.

Patient Growth Key Insights:

This page focuses on patient traffic on a monthly, daily and hourly basis in the ER alongside major key metrics. The MoM growth rate shows the rate at which the patient volume grows per month over the two year period. April had the highest growth rate of 87.5% indicating that patient volume nearly doubled. The great increase, although positive, was not consistent nor prepared for and inefficient staffing in the face of such an influx may have contributed negatively to patient experience. The overall average monthly growth rate of 8% indicates that the hospital has the potential of achieving the target of 15-20% increase in patient volume capacity however this needs to be well supported by adequate staff allocation to be achieved gradually and consistently. Focusing on patient volume, August is the busiest month - which may be linked to a general increase in hospital visits during the spring/ summer months (in the northern hemisphere) due to more outdoor activities during such seasons. Monday is the busiest day of the week with patient volume increasing from 10pm and peaking at 11pm until gradually stabilizing from 1am.

Operational Analysis by Wait Times Key Insights:

This page focused on analysing average wait times by department referral, day of week and time of day. The target wait time was determined by the brief which was 10% less than the current wait time. The overall average wait time for all patients was 35 min and therefore the target wait time was 32 minutes - 11.13% over target. Mondays and Wednesdays have the longest wait time of 36 minutes. Drilling down to the hourly analysis, patients visiting at 3am experience the longest wait time of 37 minutes which may correlate with peak patient volume between 11pm - 5am. At 12pm midday patient wait time is at the lowest of 34 minutes, similarly correlating with the reduction in patient visits at this hour while being 2 minutes from the target wait time. Patients from neurology waited the longest to be treated at 5.3 minutes over the target wait time, although these patients only account for 2% of the total patient visits.

Patient Satisfaction and Demographics

The average patient satisfaction score (sat. score) across all demographics was 5/10. The target sat. score is +1 point above the average therefore going forward the target sat. score is 6/10. However the response rate is low at only 27,33% therefore the majority of patients' satisfaction status is unknown. When focused on satisfaction by race, the most satisfied group are pacific islanders at 5,33/10 followed closely in order by native americans, african americans and asians. The least satisfied racial group are individuals of mixed ethnicity with a low score of 4.83/10 - 1,17 points shy of the target. In addition, adults had the highest satisfaction score of 5.07/10, followed by children then seniors, at 4,73/10.

The drastic drop seen in seniors may be associated with the extra care required when tending to patients older in age. The poor experience can negatively impact recommendations from patients seen at the hospital, potentially contributing to some revenue loss. Lastly, the demographics breakdown matrix combines all of the patient related analyses into one. The average satisfaction score, gap to the target score and total patient visits per department, patient ethnicity and age. By drilling down, one can get a detailed breakdown and look for correlation between patient volume and perhaps patient satisfaction as greater volume may result in longer wait times affecting patient satisfaction. For instance, adult neurology referral patients have an average sat. score of 5.11/10, lowest in comparison to children and seniors as they account for the most patient visits from the neurology department.

Recommendations

The monthly average growth rate of 8% isn't consistent, as the average is raised to 8% mainly due to the large increase in patient volume in April over the two years. Therefore aiming for 15-20% increase in patient volume capacity at this stage isn't recommended. Efforts can simultaneously be shifted to focus on first providing the necessary support that will drive consistent growth in the future. Mainly honing in on improving staffing solutions. For better staffing efficiency, prioritize having maximum manpower on Mondays, standard staffing on Tuesdays, Wednesdays and Saturdays followed by reduced staffing on Thursdays, Fridays and Sundays. Similarly on an hourly basis, maximum staff is needed between the hours of 10pm - 1am, 6am-8am and 12pm -2pm. Less staff are needed from 9am-10am and 2pm -9pm with standard staffing for the other hours of the day. The same should also be considered for spring/summer months. This will allow for staff to be utilized where necessary while cutting down on unnecessary overtime costs to support revenue growth.

Improved staffing should positively reduce wait times towards the target of 32 minutes and perhaps a deeper look into why department referrals with the fewest patient visits have some of the highest wait times i.e. neurology, physiology, gastroenterology and cardiology. Could it be related to administrative set backs, resources or availability of doctors to treat such cases? With patient satisfaction, women, individuals of mixed ethnicity and the elderly are groups that need more attention and it would be worth investigating why low satisfaction scores are being given by these individuals. This could be done by way of targeted surveys or questionnaires for these groups and from there implement the necessary improvements to reach the target satisfaction score. Finally, it is important to implement strategies to ensure more patients complete satisfaction questionnaires to increase the response rate to gather more accurate data in overall patient satisfaction, as more responses will provide greater clarity.

Methodology and data notes

This dataset was sourced from GitHub. It is synthetic and meant to mimic real life ER records.

Skills implemented below:

- Data Modelling and Cleaning:
I began with modelling the data by creating separate dimension tables and a fact table. I used the initial ER table as the fact table and created three dimension tables - DimDate for date data, DimPatients for patient information data and a DimDepartment table for department referral information. I joined the dimension tables to the fact table via unique surrogate keys creating a star schema. From there I performed simple data cleaning such as removing duplicate data, spelling corrections, grouping and formatting data.
- Calculations and DAX formulae to meet project brief KPIs:
 - Total Patient Visits =
COUNT('HospitalER'[PatientIDKey])
 - To determine the total number of patient visits (each unique patientID accounted for one visit).
 - MoM Growth Rate =
VAR CurrentMonthPatients = COUNTROWS ('HospitalER')
VAR PreviousMonthPatients = CALCULATE(COUNTROWS('HospitalER'),
DATEADD(DimDate[Date], -1, MONTH), REMOVEFILTERS(DimDate[Year]))
VAR GrowthRate = DIVIDE(CurrentMonthPatients - PreviousMonthPatients,
PreviousMonthPatients)
RETURN GrowthRate
 - Measures business expansion on a monthly basis.
 - Avg Wait Time (min) =
AVERAGE ('Hospital ER'[Wait Time Minutes])
 - Calculated to determine overall average patient wait time before receiving treatment.
 - Target Wait Time (10%) =
35.25*0.9
 - According to the project brief, the target wait time was determined to be 10% less than the current average wait time.
 - Gap to Target (min) =
VAR CurrentWT = [Avg Wait Time (min)]
VAR TargetWT = [Target Wait Time (10%)]
VAR Gap = TargetWT - CurrentWT

RETURN Gap

- Calculated to determine how many minutes need to be shaved off to reach the target wait time.
- Avg Sat. Score =
AVERAGE('Hospital ER'[Satisfaction Score])
 - The average satisfaction score submitted by patients who reported their satisfaction scores.
- Target Sat. Score =
[Avg Sat. Score] +1
 - The target satisfaction score was +1 point added to the current average satisfaction score.
- Gap to Target (score) =
[Target Sat Score] - [Avg Sat. Score]
 - Calculated to determine how many points are needed to reach the target score.
- Response Rate =
[Sat. Score Response Count]/[Total Patient Visits]
 - The response rate was calculated to observe the percentage of patients that actually gave feedback to their hospital experience out of all of the patients that visited.